FIITJEE

ALL INDIA TEST SERIES

FULL TEST – X

JEE (Advanced)-2020

PAPER – 1

TEST DATE: 20-09-2020

ANSWERS, HINTS & SOLUTIONS

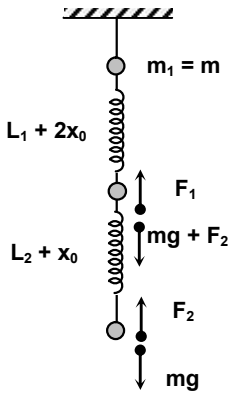
Physics

PART – I

SECTION – A

1. B

Sol. Since tensions in the 2 springs are different so they have different natural lengths. Let the natural length of upper and lower spring be L_1 and L_2 respectively.



Since, in equilibrium $\therefore F_2 = kx_2 = mg$

$\Rightarrow x_2 = \dfrac{mg}{k} = x_0 (\text{say})$

and $F_1 = mg + F_2 \Rightarrow F_1 = 2mg$

$\Rightarrow kx_1 = 2mg \Rightarrow x_1 = \dfrac{2mg}{k} = 2x_0$

Given $L_1 + 2x_0 = L_2 + x_0 \Rightarrow L_2 = L_1 + x_0$

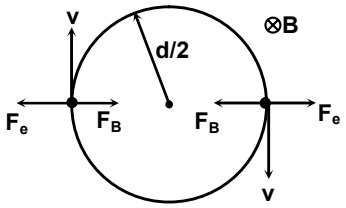
When the springs are at their natural lengths, the distance of centre of mass from top ball will be

$$x_{CM} = \frac{0 + mL_1 + m(L_1 + L_2)}{3m} = L_1 + \frac{x_0}{3}$$

$\therefore$ CM is at a distance $\dfrac{x_0}{3} = \dfrac{mg}{3k}$ below the second ball.

2. D

Sol. For distance to remain constant, electrons need to move in a circle of radius d/2.



$$\therefore\ F_B - F_e = \frac{mv^2}{d/2}$$

$$\text{If } evB - \frac{ke^2}{d^2} = \frac{mv^2}{d/2} \Rightarrow v^2 - \frac{eBd}{2m}v + \frac{ke^2}{2md} = 0$$

This is quadratic in v

$$v = \frac{edB}{4m} \pm \frac{1}{2}\sqrt{\left(\frac{edB}{2m}\right)^2 - \left(\frac{4ke^2}{2md}\right)}$$

The situation can be realized if value of v is real and positive i.e. discriminant ≥ 0

$$\Rightarrow \left(\frac{edB}{2m}\right)^2 \geq \frac{4ke^2}{2md}$$

$$\Rightarrow d_{min} = \left[\frac{2m}{\pi\varepsilon_0 B^2}\right]^{1/3} = d_{critical}$$

If $d < d_{critical}$ then problem has no solution
If $d = d_{critical}$ then problem has one solution
If $d > d_{critical}$ then problem have two solutions with different values for v.

3. C

Sol. $$F = -\frac{dU}{dr} = -2B(r - r_0)$$

Equilibrium is at $r = r_0$
When displaced by x from equilibrium position.

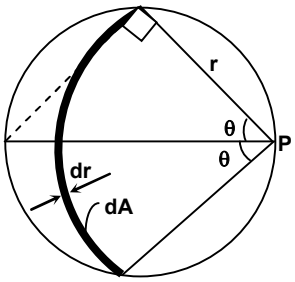
$$F_{restoring} = -2B\left[(r_0 + x) - r_0\right] = -2Bx \qquad \therefore \text{ SHM}$$

$$\text{and } a = \frac{-2B}{m_{reduced}} \cdot x = -\omega^2 x$$

where $\omega^2 = \dfrac{2B(m_1 + m_2)}{m_1 m_2}$

4. D

Sol. Work done in moving a unit positive charge from infinity to a point on the circumference will be numerically equal to the potential at that point.
So we need to calculate potential at any point on the circumference of this charged disc.
Let choose an element of area dA in the form of a ring of radius r and width dr, where $dA = 2\theta r\, dr$
$r = 2R\cos\theta \qquad \therefore\ dr = -2R\sin\theta d\theta$
Potential at point P due to the charge on this element is



$$dV = -\frac{\sigma R}{\pi\varepsilon_0}\theta\sin\theta d\theta$$

$$\therefore\ V = \int dV = -\frac{\sigma R}{\pi\varepsilon_0}\int_{\pi/2}^{0}\theta\sin\theta d\theta = -\frac{\sigma R}{\pi\varepsilon_0}\left[-\theta\cos\theta + \sin\theta\right]_{\pi/2}^{0} = \frac{\sigma R}{\pi\varepsilon_0}$$

5. A, C, D

Sol. Mass of iron block = $2 \times 10^{-3} \times 7.8 \times 10^3 = 15.6$ kg

Upthrust on iron block = $2 \times 10^{-3} \times 10^3 = 2$ kg-wt

For cube to be in equilibrium, tension in the cord should be $15.6 - 2 = 13.6$ kg-wt

And this tension should balance the weight of the water in the bucket and the reaction of the upthrust (downward on the bucket).

$\therefore$ T = wt of water + 2 (kg-wt)

$\therefore$ mass of water = $13.6 - 2 = 11.6$ kg

So volume of water is 11.6 litres

When water is poured, bucket will gradually move down and cube will rise. If additional water is more than 4 litre then weight of water in the bucket become greater than the weight of the cube.

When water in the bucket evaporates, then bucket will rise and block will gradually sink to the bottom and finally tension will be half the weight of the cube and water i.e. 7.8 kg-wt

6. B, C, D

Sol. $\Delta U_{gravitational} = 0$

So, the work done by external agent is dissipated in doing work against friction force.

$40 = 2f_rL = 2\mu mg\cos\theta\dfrac{h}{\sin\theta} = \dfrac{2\mu mgh}{\tan\theta}$

Since, block does not slide $\therefore \mu \geq \tan\theta \Rightarrow \dfrac{40\tan\theta}{2mgh} \geq \tan\theta \Rightarrow h \leq 1$ m

Now, $W_{up} = mg\sin\theta L + \mu mg\cos\theta L = mgh + \mu mgx$

$W_{down} = -mg\sin\theta L + \mu mg\cos\theta L = -mgh + \mu mgx$

$\therefore W_{total} = 2\mu mgx$

$\therefore x = \dfrac{W_{total}}{2mg\mu}$

So x is independent of θ

$\therefore$ B, C, D are correct.

7. A, D

Sol. Let length of pendulum be ℓ. For the motion along arc AP, acceleration of bob in vertical downward direction is less than equal to g. Hence the time required to cover this arc is clearly greater than it would be if the bob undergoes free fall from A to P. i.e.,

$t_1 > \sqrt{\dfrac{2\ell\sin 30^\circ}{g}} \Rightarrow t_1 > \sqrt{\dfrac{\ell}{g}}$ …(i)

Now $v_P = \sqrt{2gL\sin 30^\circ} = \sqrt{gL}$ and $v_Q = \sqrt{2gL\sin 60^\circ} = \sqrt{\sqrt{3}gL}$

Since, $v_Q > v_P$

$\therefore$ $t_3 < t_2 < t_1$ and the ball covers the arc PQ and QB with speed greater than v_P and v_Q respectively.

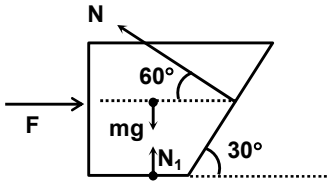
$t_{PB} < \dfrac{\ell\pi}{6v_P} + \dfrac{\ell\pi}{6v_Q} \approx 0.92\sqrt{\dfrac{\ell}{g}}$ …(ii)

Comparing (i) and (ii) $t_1 > t_{PB}$

i.e., $t_1 > t_2 + t_3$

8. B, C

Sol. $a = \dfrac{3F - F}{m + m} = \dfrac{F}{m}$ (towards left)



$N\cos 60° - F = ma = m\dfrac{F}{m}$

$\Rightarrow N \times \dfrac{1}{2} = 2F \Rightarrow N = 4F$

Also, $N \sin 60° + N_1 = mg$

$N_1 = mg - N \sin 60°$

For no slipping relative to each other, $N_1 \geq 0$

$\Rightarrow mg \geq N \sin 60° \Rightarrow mg \geq \dfrac{\sqrt{3}N}{2}$

$\Rightarrow N \leq \dfrac{2mg}{\sqrt{3}}$

$\Rightarrow N \leq \dfrac{2 \times 3 \times 10}{\sqrt{3}}$

$\Rightarrow N \leq 34.64$ Newton

9. B, C

Sol. $v = \sqrt{2gh_2}$

and for a given h_1, $v_{max} = \sqrt{2\left(\dfrac{P_{atm}}{\rho} - gh_1\right)}$

And when h_1 is reduced then v will further increase and will have the maximum possible value when $h_1 = 0$

$\therefore v_{max} = \sqrt{\dfrac{2P_{atm}}{\rho}} = 14.14$ m/s

10. A, B, C

Sol. Since, block is half submerged

$\therefore \rho_S vg = \dfrac{v}{2}\rho_L g \Rightarrow 2\rho_S = \rho_L$

(B) when temperature is decreased by ΔT (say) then fraction of volume submerged = $\dfrac{\rho_S'}{\rho_L'} = \dfrac{\rho_S}{(1-\gamma_S \Delta T)} \times \dfrac{(1-\gamma_L \Delta T)}{\rho_L} = \dfrac{(1-\gamma_L \Delta T)}{2(1-\gamma_S \Delta T)}$, as $\gamma_L > \gamma_S$ so fraction of volume submerged is decreased

(C) Let the length of the submerged part be x then

$\dfrac{x}{L} = \dfrac{\rho_S}{\rho_L}$ ...(i)

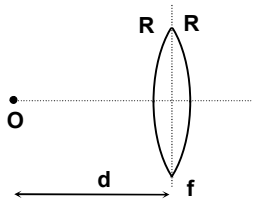
and $\dfrac{x}{L\left(1 - \dfrac{\gamma_S}{3}\Delta T\right)} = \dfrac{\rho_s(1-\gamma_L \Delta T)}{\rho_L(1-\gamma_S \Delta T)}$ ...(ii)

from (i) and (ii), $2\gamma_S = 3\gamma_L$

(D) upon heating density of liquid will increase so block will rise.

11. A, B, D

Sol. $\frac{1}{f}=\left(\frac{3}{2}-1\right)\left(\frac{1}{R}+\frac{1}{R}\right)=\frac{1}{R} \Rightarrow f=R$



$\frac{1}{f_w}=\left[\frac{3/2}{4/3}-1\right]\left(\frac{1}{R}+\frac{1}{R}\right)=\frac{1}{4f} \Rightarrow f_w=4f$

When in water: $\frac{1}{4f}=\frac{1}{v'}+\frac{1}{d} \Rightarrow \frac{1}{v'}=\frac{1}{4f}-\frac{1}{d}$

$\Rightarrow v'=\frac{4fd}{d-4f}<0$ and $\frac{v'}{u'}>1$

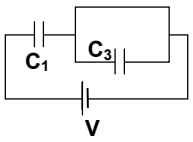
$\therefore$ Image is virtual, magnified and erect.

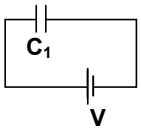
12. A, B, C

Sol. When $C_2 \to \infty \Rightarrow$ shorted

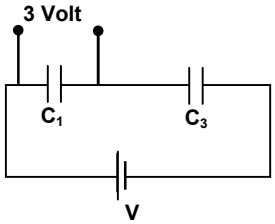
Then the circuit becomes

$\therefore V_1 = V = 12$ volt





Where $C_2 = 0$ then circuit becomes



$\therefore V_{C_3} = 12-3 = 9$ V

Since, C_1 and C_3 are in series, so charge will be same on both the capacitors

$\therefore C_1 \times 3 = C_3 \times 9 \Rightarrow \frac{C_1}{C_3}=3$

SECTION – C

13. 00002.50

Sol. For not toppling of the cylinder, $ma\frac{H}{2} \le mg\frac{H}{8} \Rightarrow a \le \frac{g}{4}$, whereas for no slipping between m and M, $a_{max}=\frac{g}{3}$.

So for equilibrium of m with respect to M, $a_{max}=\frac{g}{4}$ = 2.5 m/s^2

Now x = A sin(10t)

$\therefore \frac{d^2x}{dt^2}=a=-100A\sin(10t)$

$\therefore a_{max}$ = 100 A = 2.5

$\Rightarrow A = \frac{2.5}{100}m = 2.5$ cm

14. 00008.00

Sol. Initially $P_{in}=\frac{4T}{r}$, where P_{in} = inside pressure …(i)

$P_{outside} = 0$ (always)

When charge is sprayed, for equilibrium $P'_{in} + \frac{\sigma^2}{2\varepsilon_0} = \frac{4T}{2r}$ …(ii)

Since, isothermal process

$\therefore\ P'_{in} = \frac{P_{in}V_i}{V_f} = \frac{P_{in}}{8}$ …(iii)

Using equations (i), (ii) and (iii) we have, $\frac{4T}{8r} + \frac{\sigma^2}{2\varepsilon_0} = \frac{4T}{2r} \Rightarrow \sigma = \sqrt{\frac{3T\varepsilon_0}{r}}$

$\therefore\ Q = 4\pi(2r)^2\sigma = 16\pi r^2\sqrt{\frac{3T\varepsilon_0}{r}} = 8\sqrt{12\pi^2 r^3 T\varepsilon_0}$

15. 00002.20

Sol. $E = \sigma T^4$

$\Rightarrow \frac{E_2}{E_1} = \frac{T_2^4}{T_1^2} = 256 = 4^4$

$\Rightarrow T_2 = 4T_1$

Also, $\lambda_1 T_1 = \lambda_2 T_2$ gives $\lambda_2 = \frac{\lambda_1}{4} = 1000\ \text{Å}$

$KE_{max} = \frac{hC}{\lambda_2} - \phi \Rightarrow \frac{12400}{1000} - \phi = 12.4 - \phi$ …(i)

$\Delta E_{2\to4} = 13.6 \times 2^2 \times \left[\frac{1}{2^2} - \frac{1}{4^2}\right] = 13.6 \times 4 \times \frac{3}{16} = 10.2\ \text{eV}$ …(ii)

From (i) and (ii) $12.4 - \phi = 10.2$

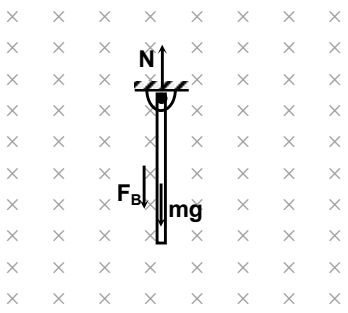
$\Rightarrow \phi = 2.2$ eV

16. 00032.50

Sol. Loss in gravitational potential energy = Gain in kinetic energy

$Mg\left[\frac{L}{2} - \frac{L}{2}\cos 60°\right] = \frac{1}{2}\frac{ML^2}{3}\omega^2$

$\omega = \sqrt{\frac{3g}{2L}} = \sqrt{15}$ rad/s



When rod is at its lowest position the forces acting on it are as shown in the diagram.

Magnetic force on the rod

$F_B = \int dq(\vec{v} \times \vec{B}) = \int_{x=0}^{x=L}\left(\frac{q}{L}dx\right) \times (\omega x) \times B = \frac{1}{2}qBL\omega$

$\therefore\ N - mg - F_B = m\omega^2\frac{L}{2}$

$N = mg + F_B + m\omega^2\frac{L}{2}$

$= 1 \times 10 + \frac{1}{2} \times 1 \times \sqrt{15} \times 2\sqrt{15} \times 1 + 1 \times (\sqrt{15})^2 \times \frac{1}{2}$

$= 10 + 15 + 7.5 = 32.5$ N

17. 00001.25

Sol. Let T be the tension in the wire at the middle. Considering rotational equilibrium of the half wire about fixed end.

Torque due to tension = torque due to its weight

$$Td = \mu\frac{L}{2}g\frac{L}{4}$$

$$\Rightarrow T = \frac{\mu L^2}{8d}g$$

$$\therefore v = \sqrt{\frac{T}{\mu}} = L\sqrt{\frac{g}{8d}} = 10\sqrt{\frac{10}{8\times 8\times 10^{-3}}} = 125 \text{ m/s}$$

$= 1.25 \times 10^2$ m/s

18. 00989.80

Sol. Since potential difference across all the inductors are same

$$\text{So, } L_1\frac{di_1}{dt} = L_2\frac{di_2}{dt} = L_3\frac{di_3}{dt}$$

$\Rightarrow i_1 = i_2 + C_2 = i_3 + C_3$

$\Rightarrow 1 = 2 + C_2 = 4 + C_3$

$\Rightarrow C_2 = -1$ and $C_3 = -3$

$$\text{Also } L_1\frac{di_1}{dt} + \frac{q}{C} = 0 \quad \text{or} \quad \frac{L}{3}\frac{d^2q}{dt^2} + \frac{q}{C} = 0 \quad \text{or} \quad \frac{d^2q}{dt^2} = -\left(\frac{3}{LC}\right)q = -\omega^2 q$$

$\therefore q = q_{max} \sin \omega t$

and $i = q_{max}\omega \cos \omega t$

at $t = 0$, $i = 1 + 2 + 4 = 7$A

$\therefore 7 = q_{max}\omega$

$$\Rightarrow q_{max} = \frac{7}{\omega} = 7\sqrt{\frac{LC}{3}} = 989.8\times 10^{-6}\text{C} = 989.8\ \mu\text{C}$$

Chemistry

PART – II

SECTION – A

19. A

Sol. Since closed chamber is there, hence there will be an equilibrium. So relative lowering of vapour pressure will be same.

$$\chi_{C_6H_{12}O_6} = \chi_x$$

$$\frac{40/180}{\frac{40}{180}+\frac{60}{18}} = \frac{10/M_x}{\frac{10}{M_x}+\frac{90}{18}}$$

$$\frac{4/18}{\frac{4}{18}+\frac{60}{18}} = \left(\frac{10/M_x}{\frac{10}{M_x}+5}\right)$$

$$\Rightarrow \frac{4}{64} = \frac{1}{16} = \frac{10/M_x}{\left(\frac{10}{M_x}+5\right)}$$

$$\frac{10}{M_x}+5 = \frac{16\times 10}{M_x}$$

$$\frac{15\times 10}{M_x} = 5$$

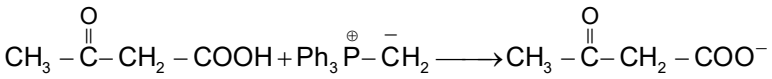
$$M_x = 30$$

20. B

Sol. Order of K_a values $\underset{10^{-9}}{y} > \underset{10^{-13}}{z} > \underset{10^{-20}}{x}$

21. C

Sol.



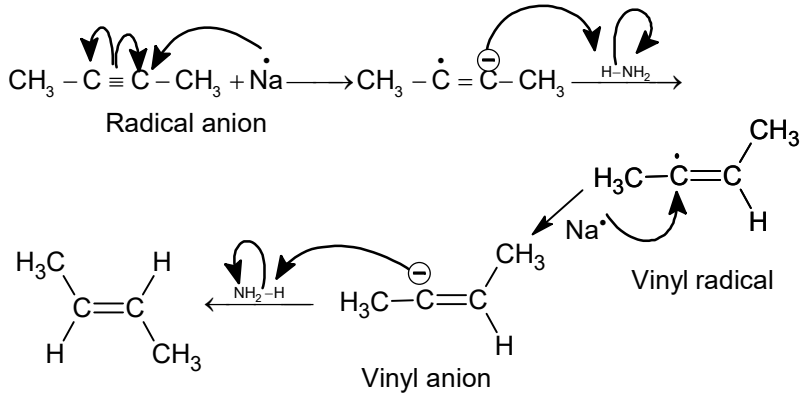
$Ph_3\overset{\oplus}{P}-\overset{-}{C}H_2$, it's one equivalent work as base.

22. B

Sol. - NH_2 is 1° amine and in rest of option lone pair of star (*) marked nitrogen is taking part in resonance. Or sp^2 hybridized.

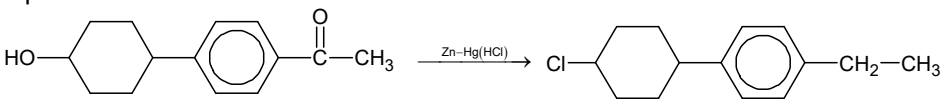
23. A, B, C, D

Sol. - 78°C temperature is required for above reaction.



24. B, C, D

Sol. Option A is incorrect. Rest are correct.



25. C, D

Sol. Lactose is present in milk.

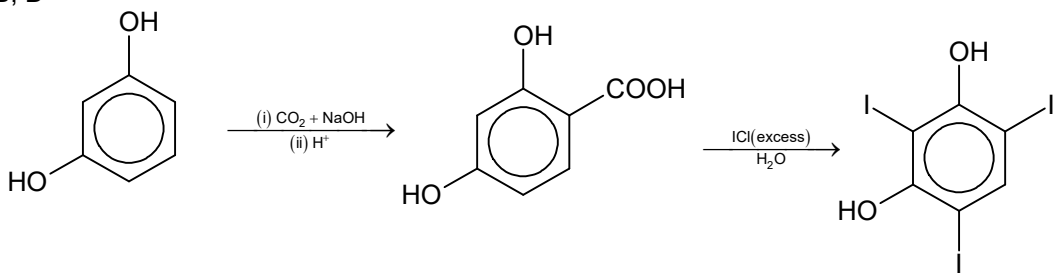
Maltose formed by α-D glucose and α-D glucose.

Hydrolysis of sugar change dextro to laevo.

26. B, C, D

27. B, D

Sol.



28. A, B, C, D

Sol. Mirror images are non super-imposable.

29. A, B, C

Sol. D is incorrect.

$$CrO_2Cl_2 + 4NaOH \longrightarrow 2NaCl + Na_2CrO_4 + 2H_2O$$

30. A, B, C, D

SECTION – C

31. 00001.25

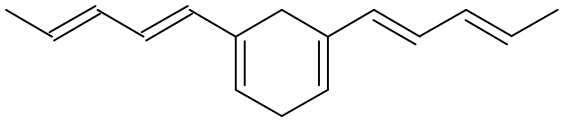
Sol. Emulsion $\rightarrow$ Butter, milk, blood, shampoo, polish, x = 5

Gels $\rightarrow$ Cheese, curd, jam, jellies, y = 4

$$\frac{x}{y}=\frac{5}{4}=1.25$$

32. 00004.50

Sol.



$$\xrightarrow[(ii)\ H_2O+Zn]{(i)\ O_3} 2CH_3CHO + \begin{matrix}2\,CHO\\ |\\ CHO\end{matrix}$$

$$+\ OHC-\overset{\overset{O}{\|}}{C}-CH_2-\overset{\overset{O}{\|}}{C}-CHO$$

$$+\ CHO-CH_2-CHO$$

a=36, b= 6 then 36/(6+2) = 4.50

33. 00002.50

Sol. a, c, d, e, f are compounds which give Reimer Tiemann and other do not give.

So 5/2 = 2.5

34. 00001.20

Sol. $3Cl_2+6NaOH(conc.)\longrightarrow 5NaCl+NaClO_3+H_2O$

$3Br_2+6NaOH(conc.)\longrightarrow 5NaBr+NaBrO_3+3H_2O$

$3I_2+6NaOH(conc.)\longrightarrow 5NaI+NaIO_3+3H_2O$

$3H_2O+P_4+Conc.3NaOH\longrightarrow PH_3+3NaH_2PO_2$

$(CH_3)_2CHCHO+Conc.NaOH\longrightarrow (CH_3)_2CHCOO^-Na^+ +(CH_3)_2CHCH_2OH$

$HCHO+Conc.NaOH\longrightarrow HCOO^-Na^+ +CH_3OH$

Total 6 compound undergo disproportionation.

So x= 6 & y= 5 x/y=1.2

35. 00012.50

Sol.

$$7A\longrightarrow B$$

$t=0$ a 0

$t=15hr$ $a-b$ $\frac{b}{7}$

for point P $\Rightarrow a-b=\frac{b}{7}\Rightarrow a=\frac{8b}{7}$

$$b=\frac{7a}{8}$$

then $a-b\Rightarrow a-\frac{7a}{8}=\frac{a}{8}$

$$a\xrightarrow{t_{1/2}}\frac{a}{2}\xrightarrow{2t_{1/2}}\frac{a}{4}\xrightarrow{3t_{1/2}}\frac{a}{8}$$

3 half lives will be 900 hrs

$3t_{1/2} = \frac{900}{24}$ days

$t_{1/2}$ = 12.5 days

36. 00003.50

Sol. $x \to 3$

$y \to 4$

$$\frac{x+y}{2} = \frac{3+4}{2} = 3.50$$

Compounds

Chile salt petre $\to NaNO_3$, Lunar caustic $\to AgNO_3$, Borax $\to Na_2B_4O_7.10H_2O$,

Milon's base $\to NH_2HgOHgI$

Heating effect

$$2Ag_2CO_3 \xrightarrow{\Delta} 4Ag + 2CO_2 + O_2$$

$$(NH_4)_2CO_3 \xrightarrow{\Delta} NH_4HCO_3 + NH_2COONH_4 + H_2O + NH_3$$

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$

$$2NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$$

$$2Hg_2CO_3 \longrightarrow 4Hg + 2CO_2 + O_2$$

Mathematics

PART – III

SECTION – A

37. B

Sol. $f^2(x) + xf(x) = 3 \Rightarrow f'(x) = \dfrac{-f(x)}{x + 2f(x)}$

$$\Rightarrow \int \frac{3x^3 + 6x^2 f(x) + 2f(x)}{(x + 2f(x))(x^3 - 2f(x))^2}\,dx = \int \frac{3x^2(x + 2f(x)) + 2f(x)}{(x + 2f(x))(x^3 - 2f(x))^2}\,dx$$

$$\Rightarrow \int \frac{3x^2 - 2f'(x)}{(x^3 - 2f(x))^2}\,dx = \frac{1}{2f(x) - x^3} + c$$

38. D

Sol. Required probability = $\dfrac{1}{2} + \dfrac{1}{2}\left(\dfrac{1}{2}\cdot\dfrac{1}{2} + \dfrac{1}{2}\cdot\dfrac{1}{2}\cdot\dfrac{1}{2}\cdot\dfrac{1}{2}\cdot\dfrac{1}{2}\cdot\dfrac{1}{2}\cdot\dfrac{1}{2}\cdot\dfrac{1}{2}\cdot\dfrac{1}{2}\right) = \dfrac{22}{32}$

39. A

Sol. Required probability = $\dfrac{\dfrac{{}^5C_2}{x}}{\dfrac{{}^5C_2 + {}^5C_3}{x}} = \dfrac{1}{2}$

40. B

Sol. $D = (2n + 1)^2 - 4(2m + 1)(2p + 1)$

$\Rightarrow$ Roots are rational if D is perfect square.

$\Rightarrow (2n + 1)^2 - 4(2m + 1)(2p + 1) = I^2$ (I = Integer)

$\Rightarrow (2n + 1)^2 - (2r + 1)^2 = 4(2m + 1)(2p + 1)$, $I = 2r + 1$, r = Integer

$\Rightarrow 4(n + r + 1)(n - r) = 4(2m + 1)(2p + 1)$

$\Rightarrow (n + r + 1)(n - r) =$ odd (not possible)

Hence, roots are irrational.

41. B, D

Sol. $\Rightarrow \tan^{-1}\left(\dfrac{r^2 + 2r + 1 - r^2}{1 + r^2(r^2 + 2r + 1)}\right)$

$$\Rightarrow \tan^{-1}\left(\frac{(r+1)^2 - r^2}{1 + r^2(r+1)^2}\right) = \tan^{-1}(r+1)^2 - \tan^{-1}(r^2)$$

42. B, D

Sol. $L = \lim\limits_{n\to\infty}\left(1 + \dfrac{1}{n}\right)^n\left(1 + \dfrac{1}{2n}\right)^n\left(1 + \dfrac{1}{2^2 n}\right)^n \ldots\left(1 + \dfrac{1}{2^{n-1}\cdot n}\right)^n$

$\Rightarrow L = e^{1 + \frac{1}{2} + \frac{1}{4} + \ldots} \Rightarrow e^2 \Rightarrow |k| = 2 \Rightarrow k = \pm 2$

43. A, B, C

Sol. $\Rightarrow \left(\sum\limits_{r=1}^{n}\left(r\sum\limits_{p=1}^{r}(\omega^{p-1})\right)\right) - 155\omega$ is real

$\Rightarrow$ 2 + 5 + 8 + ….. k terms = 155 $\Rightarrow$ k = 10
Hence, n can be 29, 30, 31 only

44. A, C

Sol. $y = \lambda_1 \cos 2x + \lambda_2$

$\Rightarrow$ Differential equation is $(\sin 2x)\dfrac{d^2y}{dx^2} = 2 \cdot \dfrac{dy}{dx} \cdot \cos 2x$

$\Rightarrow \lambda = 0, f(x) = \sin 2x$

45. A, C

Sol. Put $\sec^2 x = t$

$\Rightarrow$ The given expression will be converted to $\sqrt{t-4} + 2\sqrt{6-t}$

$\Rightarrow$ Minimum value = $\sqrt{2}$, Maximum value = $\sqrt{10}$

46. A, B, C

Sol. $\dfrac{\sin(x+a)}{\sin(x+b)} = \cos(a-b) + \cot(x+b)\sin(a-b)$

$\Rightarrow$ If $a - b = n\pi$, $n \in I$ then the function is constant

47. A, B, C

Sol. $A^3 = \begin{bmatrix} 29 & 35 & 45 \\ 45 & 54 & 70 \\ 25 & 30 & 39 \end{bmatrix}$

48. B, D

Sol. Let the pair of lines be $(y = mx + c_1)(y - m^2x + c_2) = 0$

$\Rightarrow a = m^3$ and $m^2 + m = 6$

$\Rightarrow m = -3, 2 \Rightarrow a = -27, 8$

SECTION – C

49. 00002.00

Sol. $f(k) = \dfrac{k^2(k^2+1)}{2}$, $g(k) = \left(\dfrac{k(k+1)}{2}\right)^2 \Rightarrow \lim\limits_{x\to\infty} \dfrac{f(x)}{g(x)} = 2$

50. 00007.00

Sol. $\alpha = \lim\limits_{k\to\infty} \sum\limits_{r=1}^{2k} \dfrac{k+2r}{k^2+kr+r^2} \Rightarrow \alpha = \lim\limits_{k\to\infty} \sum\limits_{r=1}^{2k} \left(\dfrac{1+2\cdot\left(\frac{r}{k}\right)}{1+\left(\frac{r}{k}\right)+\left(\frac{r}{k}\right)^2} \right)$

$\Rightarrow \alpha = \int\limits_0^2 \dfrac{1+2x}{1+x+x^2} dx \Rightarrow \alpha = \ln 7$

51. 00002.00

Sol. $f(x) = \dfrac{x}{1+x^2} \Rightarrow \int\limits_{-\pi/4}^{\pi/4} \dfrac{f(x)+x^2-x^3+x+1}{\cos^2 x} dx = \int\limits_{-\pi/4}^{\pi/4} \sec^2 x \, dx = 2$

52. 00256.00

Sol. $m = 0, n = -4 \Rightarrow m^4 + n^4 = 256$

53. 00000.00

Sol. $y = \left(\frac{8x^3}{7-6x^4}\right)^{\frac{1}{3}} \Rightarrow \lim_{x\to\infty} y = 0$

54. 00001.57

Sol. Area bounded = $2\left[\frac{1}{2}\cdot 2^2\cdot\frac{\pi}{6} - \frac{1}{2}\cdot 1^2\cdot\frac{\pi}{6}\right] \Rightarrow$ Area = $\frac{\pi}{2} = 1.57$